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Textbooks Are All You Need II: **phi-1.5** technical report

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Abstract

We continue the investigation into the power of smaller Transformer-based language models as initiated by **TinyStories** – a 10 million parameter model that can produce coherent English – and the follow-up work on **phi-1**, a 1.3 billion parameter model with Python coding performance close to the state-of-the-art. The latter work proposed to use existing Large Language Models (LLMs) to generate "textbook quality" data as a way to enhance the learning process compared to traditional web data. We follow the "Textbooks Are All You Need" approach, focusing this time on common sense reasoning in natural language, and create a new 1.3 billion parameter model named **phi-1.5**, with performance on natural language tasks comparable to models 5x larger, and surpassing most non-frontier LLMs on more complex reasoning tasks such as grade-school mathematics and basic coding. More generally, **phi-1.5** exhibits many of the traits of much larger LLMs, both good –such as the ability to "think step by step" or perform some rudimentary in-context learning– and bad, including hallucinations and the potential for toxic and biased generations –encouragingly though, we are seeing improvement on that front thanks to the absence of web data. We open-source **phi-1.5** to promote further research on these urgent topics.

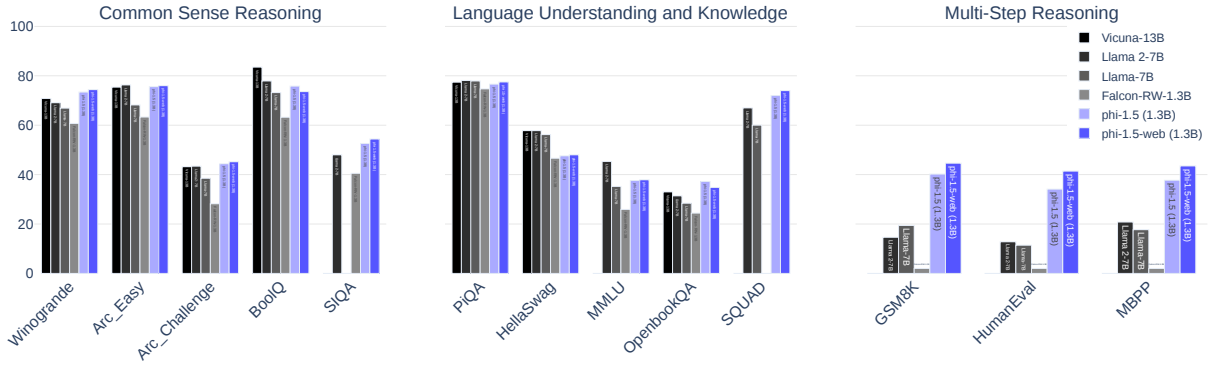

Figure 1: Benchmark results comparing **phi-1.5**, its version enhanced with filtered web data **phi-1.5-web**, and other state-of-the-art open-source LLMs. Sizes range from **phi-1.5**'s 1.3 billion parameters (Falcon-RW-1.3B [PMH+23]) to 10x larger models like Vicuna-13B [ZCS+23], a fine-tuned version of Llama-13B [TLI+23]). Benchmarks are broadly classified into three categories: common sense reasoning, language skills, and multi-step reasoning. The classification is meant to be taken loosely, for example while HellaSwag requires common sense reasoning, it arguably relies more on "memorized knowledge". One can see that **phi-1.5** models perform comparable in common sense reasoning and language skills, and vastly exceeds other models in multi-step reasoning. Note that the numbers are from our own evaluation pipeline, to ensure consistency between models, and thus they might differ slightly from numbers reported elsewhere.

1 Introduction

Over the past few years, Large Language Models (LLMs) have transformed the field of Natural Language Processing. More broadly, they hold the promise of a paradigm shift for human-computer interaction. These advancements have far-reaching economic implications, as well as the potential to redefine our conceptual frameworks of artificial intelligence and perhaps even cognition itself. Moreover, the latest generation of models such as GPT-4 [Ope23] have demonstrated remarkable improvements over their predecessors, offering capabilities previously thought to be unattainable in the short term; see for example [BCE+23] for an in-depth comparison between GPT-4 and its predecessor GPT-3.5.

The improvement from one generation of LLMs to the next seems at the moment to primarily stem from *scale*, with the most powerful models nearing trillions of parameters and trillion of tokens for training data (for example, PaLM [CND+22] has 540 billion parameters and was trained on 780 billion tokens). A natural question arises: Is this large scale indispensable for achieving high levels of capability? Far from being merely an academic question, answering this holds implications across several dimensions. Economically, the cost of training, deploying, and maintaining such large models can be substantial. Scientifically, understanding whether similar capabilities can be achieved at a smaller scale could provide insights into the architectures and development of intelligent systems. From a responsible AI standpoint, the energy consumption of large-scale models is becoming an increasing concern, as is the question of how controllable or governable these large models can be. Finally, the ability to train compact models with cutting-edge capabilities would democratize advanced AI, enabling a broader range of individuals and organizations to study and deploy them, instead of being an exclusive domain of a few with vast computational resources.

In this work we continue the investigation into the fundamental question of "how small can a LLM be to achieve certain capabilities". The prior work [EL23] considered this question for the task of "speaking fluent English", while the subsequent work [GZA+23] considered the more challenging task of coding simple functions in Python. Here we focus on the more elusive concept of *common sense reasoning*, a notoriously challenging task for AI [SBBC21]. Our results are summarized in Figure 1. In a nutshell we build **phi-1.5**, a 1.3 billion parameter model trained on a dataset of 30 billion tokens, which achieves common sense reasoning benchmark results comparable to models ten times its size that were trained on datasets more than ten times larger. Moreover, our dataset consists almost exclusively of synthetically generated data (closely following the approach from [GZA+23], see next section for more details), which has important implications for the potential to control for the notoriously challenging issue of toxic and biased content generation with LLMs [BGMMS21]. Additionally, we discuss the performance of a related *filtered web data* enhanced version of **phi-1.5**, which we call **phi-1.5-web** .

We open-source our raw **phi-1.5** model (without instruction fine-tuning or any other stage of alignment) to empower the research community in its work on some of the most urgent questions around LLMs: in-context learning, mechanistic interpretability, and mitigation strategies for hallucinations, toxic content generation, and biased outputs. Indeed, **phi-1.5** is the first LLM at the one billion parameters scale to exhibit most of the relevant traits of larger LLMs for research on these topics. We hope that **phi-1.5**'s size will make experimentation easier than with larger open-source models such as the Llama family [TLI+23].

	Train time (GPU hrs.)	MicroBatch (max)	Inf. speed (per token)	Inf. memory (at 2048 ctx.)	Data size (tokens)	Train tokens
Llama-7B	> 80K	2	14ms	18G	1T	1T
phi-1.5 (1.3B)	1.5K	8	<3ms	3.5G	30B	150B
phi-1.5-web (1.3B)	3K	8	<3ms	3.5G	100B	300B

Table 1: Comparison of compute of different models using a single A100-80G with context length 2048 and fp16.

2 Technical specifications

We give here details of the creation of **phi-1.5**. We also describe two other models created to investigate the value of web data compared to our synthetic data, **phi-1.5-web-only** and **phi-1.5-web**.

2.1 Architecture

The architecture for **phi-1.5** (and its variants) is exactly the same as our previous model **phi-1** in [GZA+23]. It is a Transformer [VSP+17] with 24 layers, 32 heads, and each head has dimension 64. We use rotary embedding with rotary dimension 32, and context length 2048. We also use flash-attention [DFE+22, Dao23] for training speed up, and we use the tokenizer of codegen-mono [NPH+22].

2.2 Training data

Our training data for **phi-1.5** is a combination of **phi-1**'s training data (7B tokens) and newly created synthetic, "textbook-like" data (roughly 20B tokens) for the purpose of teaching common sense reasoning and general knowledge of the world (science, daily activities, theory of mind, etc.). We carefully selected 20K topics to seed the generation of this new synthetic data. In our generation prompts, we use samples from web datasets for diversity. We point out that the only non-synthetic part in our training data for **phi-1.5** consists of the 6B tokens of filtered code dataset used in **phi-1**'s training (see [GZA+23]).

We remark that the experience gained in the process of creating the training data for both **phi-1** and **phi-1.5** leads us to the conclusion that the creation of a robust and comprehensive dataset demands more than raw computational power: It requires intricate iterations, strategic topic selection, and a deep understanding of knowledge gaps to ensure quality and diversity of the data. We speculate that the creation of synthetic datasets will become, in the near future, an important technical skill and a central topic of research in AI.

2.3 Training details

We train **phi-1.5** starting from random initialization with constant learning rate $2e-4$ (no warm up)[1], weight decay 0.1. We use Adam optimizer with momentum $0.9, 0.98$, and epsilon $1e-7$. We use fp16 with DeepSpeed ZeRO Stage 2 [RRRH20]. We use batch size 2048, and train for 150B tokens, with 80% from the newly created synthetic data and 20% from **phi-1** 's training data.

2.4 Filtered web data

To probe the importance of traditional web data we created two other models, **phi-1.5-web-only** and **phi-1.5-web**. To do so we create a dataset of 95B tokens of *filtered web data* following the filtering technique in [GZA+23]. This *filtered web data* consists of 88B tokens filtered from the Falcon refined web dataset [PMH+23], and 7B tokens of code data filtered from The Stack [KLA+22] and StackOverflow.

Our **phi-1.5-web-only** model is trained purely on the *filtered web data* with about 80% training tokens from NLP data sources and 20% from code datasets (no synthetic data). Our **phi-1.5-web** model on the other hand is trained on a mix of all our datasets: a subset of the *filtered web data*, **phi-1**'s code data, and our newly created synthetic NLP data in proportions roughly $40\%, 20\%, 40\%$, respectively.

Remark: None of our models have undergrone instruction finetuning or RLHF. Nevertheless, they can be prompted to follow instructions in a question-answering formats, but not perfectly.

[1] The training configuration is intentionally kept straightforward to emphasize the significance of our data.